

RUT Stage B Preliminary Results

Growth / RSD / Structure Compatibility Test for the Frozen M1a Late-Time Bridge

Prepared for GitHub / OSF documentation and replication packaging

Status: Preliminary. Scope: first-pass RSD/fsigma8 growth testing using the Stage A frozen M1a bridge. This report is not a proof of RUT; it documents whether the Stage A distance-map bridge survives growth pressure.

1. Executive conclusion

Stage B gives a clean but modest compatibility result. The Stage A M1a bridge is not rejected by growth. Across smoke tests, leave-one-out checks, parameter-symmetry checks, S8 health diagnostics, low-vs-high redshift splits, prior stress tests, and gamma-index diagnostics, the frozen M1a bridge remains either weakly favored or tied with LCDM. The strongest Stage B statement is not that RUT dominates growth; it is that the Stage A bridge survives growth pressure without requiring obviously pathological GR-like growth behavior.

Reviewer-safe headline: The Stage A M1a bridge remains growth-compatible. In current RSD/fsigma8 smoke tests, M1a is mildly BIC-favored in matched comparisons, physically healthy in S8 for the frozen solution, mostly supported at low redshift, and nearly standard in its GR-like effective growth index.

2. Frozen model and dataset

Stage B used the Stage A frozen M1a bridge values, rather than re-tuning the bridge from scratch. This was intentional: the test asks whether the same bridge that improved Stage A transverse BAO distance behavior also survives growth and structure checks.

Parameter	Frozen Stage A value used in Stage B
z_t	0.35
width	0.03
dG	-0.159
H0	73.6
Omega_m	0.248
r_d	146.7

The RSD/fsigma8 file contained six rows: Main Galaxy Sample ($z=0.135$), BOSS Galaxy ($z=0.350$), BOSS Galaxy.1 ($z=0.500$), eBOSS LRG ($z=0.800$), eBOSS ELG ($z=0.850$), and eBOSS Quasar ($z=1.500$).

3. Test-by-test summary

Step	Question	Key result	Interpretation
B1	Did the Stage B setup load correctly?	6 RSD/fsigma8 rows loaded; Stage A M1a values frozen.	Clean setup.
B2	Does frozen M1a survive first growth smoke test?	M1a frozen GR growth: $\chi^2=4.525$, $BIC=6.317$, ΔBIC vs LCDM=-2.780.	Moderate first-pass support.
B3	Is the growth result driven by one point?	6/6 leave-one cases favored M1a by $\Delta BIC < 0$; 4/6 by $\Delta BIC < -2$.	Modest but stable.
B4	Does support survive fair parameter matching?	Matched $k=1$ $\Delta BIC=-1.048$; matched $k=2$ $\Delta BIC=-1.689$.	Weak but persistent support.
B5	Are parameters healthy?	Best model M1a fixed Ω_m : $S8=0.7895$; health flag healthy_first_pass.	No obvious S8 problem for frozen model.
B6	Which points improve?	Fixed- Ω_m M1a improved 5/6 points; strongest improvement BOSS $z=0.350$; worsened eBOSS Quasar $z=1.500$.	Low-redshift support; high-z pressure.

Step	Question	Key result	Interpretation
B7	Is support low-z or high-z?	Low-z Delta BIC=-1.078; high-z Delta BIC=0.000.	Growth support concentrated at low redshift.
B8	Does support survive Omega_m/S8 priors?	Fixed-Omega_m M1a stayed weakly favored under every prior stress case, Delta BIC about -1.05 to -1.07.	Stable and healthy.
B9	Does growth require strange gamma?	M1a gamma fixed Omega_m: Delta BIC=-1.881; GR-like gamma_eff mean=0.5537.	No weird GR-like growth behavior.

4. Core smoke test results

B2 gave the first answer: frozen M1a with GR-like growth performed better than the LCDM growth fit by BIC. The extra μ_0 correction barely reduced chi-square and was penalized by BIC, so it should not be emphasized at this stage.

Model	chi2	BIC	Delta BIC vs LCDM	Parameters
M1a frozen GR growth	4.5249	6.3166	-2.7795	sigma8=0.8684
M1a frozen μ_0 growth	4.5191	8.1026	-0.9936	sigma8=0.8708, μ_0 =0.1319
LCDM free Ω_m +sigma8	5.5127	9.0962	0.0000	Omega_m=0.2274, sigma8=0.8713

5. Robustness and fairness

B3 showed that the growth support was not caused by a single RSD row. In the leave-one-out run, 6/6 cases still favored M1a over LCDM by Delta BIC < 0, and 4/6 cases favored M1a by Delta BIC < -2. B4 then made the claim more modest: once parameter freedom was matched, M1a remained favored, but only weakly.

Comparison	Delta BIC	Best M1a parameters	Health read
Matched k=1 fixed Omega_m	-1.048	Omega_m=0.248, sigma8=0.8684, S8=0.7895	healthy_first_pass
Matched k=2 free Omega_m	-1.689	Omega_m=0.1881, sigma8=0.9118, S8=0.7219	low_Omega_m warning

6. Parameter health

The healthiest and best-BIC Stage B model remained M1a with Omega_m fixed to the Stage A value. It gave Omega_m=0.248, sigma8=0.8684, and S8=0.7895. This was flagged as healthy_first_pass. The free-Omega_m M1a variants could improve chi-square but tended to drift into less clean parameter territory, especially low Omega_m in B5 or high gamma/S8 in the free gamma diagnostic. Therefore, the fixed-Omega_m frozen M1a solution is the clean version to emphasize.

7. Redshift localization

B6 and B7 show that the growth support is concentrated at low redshift. The primary fixed-Omega_m residual decomposition found that M1a improved 5/6 RSD points. The top improvement was BOSS Galaxy $z=0.350$, which is notable because this is the same redshift neighborhood that mattered in Stage A. The one worsened point was eBOSS Quasar $z=1.500$. The redshift split then showed low-z support and high-z tying.

Window	Matched k=1 Delta BIC	Matched k=2 Delta BIC	Read
Full all	-1.048	-1.689	Weakly favors M1a.
Low $z < 0.6$	-1.078	-0.168	Low- z growth support.
High $z \geq 0.6$	0.000	-0.000	Tied with LCDM.

8. Prior stress

B8 checked whether the Stage B result collapses under soft Ω_m or S8 health priors. The frozen fixed- Ω_m M1a model was stable under every prior stress case and remained weakly favored relative to the matched LCDM fixed- Ω_m model. This is a useful hardening result because it shows that the clean Stage B model is not relying on low- Ω_m drift.

Prior stress case	Fixed- Ω_m Delta BIC	M1a Ω_m	M1a S8	Health
No prior	-1.048	0.248	0.7895	healthy
Stage A Ω_m prior	-1.048	0.248	0.7895	healthy
Broad $\Omega_m=0.30$ prior	-1.048	0.248	0.7895	healthy
Broad S8 prior	-1.072	0.248	0.7897	healthy
Ω_m +S8 prior	-1.072	0.248	0.7897	healthy

9. Gamma-index diagnostic

B9 tested whether M1a requires strange growth-index behavior. The gamma-fit version with fixed Ω_m weakly favored M1a. More importantly, the GR-like effective gamma from the actual growth solution remained close to standard behavior: mean $\gamma_{\text{eff}}=0.5537$ for M1a versus 0.5499 for LCDM. That is a strong health check. It suggests the frozen bridge modifies low-redshift behavior without forcing abnormal GR-like growth.

Diagnostic	M1a	LCDM	Read
Gamma fixed Ω_m	BIC=7.233, $\gamma=0.7280$, S8=0.8309	BIC=9.114, $\gamma=0.5603$	M1a weakly favored; healthy first pass.
Gamma free Ω_m	BIC=8.853, $\gamma=1.2000$, S8=1.0213	BIC=9.948, $\gamma=0.3222$, S8=0.5202	Do not emphasize; both free fits drift.
GR-like effective gamma	mean=0.5537, range=0.5468-0.5643	mean=0.5499, range=0.5468-0.5541	M1a GR-like growth stays close to standard.

10. Caveats

- Stage B uses only six RSD/fsigma8 points, so the evidence is preliminary and should not be overclaimed.
- The growth evidence is weaker than Stage A's transverse-distance signal.
- The high-redshift growth window is tied with LCDM, not better than LCDM.
- The free- Ω_m and free-gamma variants can drift into uncomfortable parameter regions and should not be the lead claim.
- This stage does not yet include covariance-aware growth/lensing constraints, a full E_G statistic, weak-lensing shear, or a Stage C joint background-plus-growth fit.

11. Recommended Stage C transition

Stage B is strong enough to freeze as a preliminary growth-compatibility pass. The next necessary test is Stage C: a joint consistency fit combining the Stage A background lane with the Stage B growth lane. Stage C should ask whether the same M1a bridge remains favored when BAO, $H(z)$, RSD/ $f\sigma_8$, H_0 prior, parameter health, and BIC/AIC accounting are evaluated together.